MIR MUNAVVAR ALI

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SUMMARY : Mechanical Design Engineer with hands on experience of developing electromechanical hardware for robotics industrial applications. Experienced in CAD/CAE, FEA, sheet metal design, structural and thermal analysis, additive manufacturing, & hands-on MATLAB-based controls. Proven ability to solve design problems, optimize hardware performance, & improve product reliability

EDUCATION

Northeastern University **Boston, USA**

Master's in Mechanical Engineering with Concentration in Mechatronics *Jan 2025 - May 2027*

- Relevant coursework: Control Systems, Mobile Robotics, Mechatronics systems, MEMS , Energy storage & Harvesting

Osmania University **Hyderabad, IND**

Bachelor's in Mechanical Engineering *Jul 2018 - Jun 2022*

- Relevant coursework: CAD/CAM, Machine design, Strength of Materials, Basics of Electronics, FEA, DFM/DFA, GD&T

INDUSTRY EXPERIENCE

Mechanical Technologist Engineer - Co-op | Veeco Instruments Inc | Plainview, NY, USA **Jan 2026 - Aug 2026**

- Cut neutralizer component size by 2× and doubled operational lifetime, as measured by bench testing against the prior design, by running iterative prototype and test cycles on a $5M+ Ion Beam Deposition program.
- Shortened prototype iteration by 3D-printing alumina ceramic and semiconductor-grade parts and designing impedance-matching networks for compact vacuum sub-assemblies.
- Resolved recurring mechanical design issues by performing root-cause analysis and redesigning sheet metal components to improve reliability. Released drawings to ASME/ANSI Y14.5, BOMs, and ECOs in SAP/ECTR (PLM)

Junior Mechanical Design Engineer | Vir Innovations Pvt Ltd | HYD, IND **Mar 2023 - Dec 2024**

- Cut prototype iteration time 33% (12 to 8 weeks) by validating chassis and actuator mounts for a stair-climbing robot and a vehicle-alerting system with static and dynamic FEA in SolidWorks; confirmed servo integration with Arduino.
- Qualified IP-rated housings for water, corrosion, and thermal protection with zero electrical failures after sand, water, and moisture exposure tests; ran environmental validation and design reviews supporting patent and product development.

Mechanical Design Engineer - Intern | Mowito Automation | PUN, IND **Nov 2022 - Dec 2022**

- Sized a linear actuator for a 15 kg load at 33 mm/s (screw life, load capacity, motor power) in SolidWorks/Onshape; verified brake margin (17 N·m required vs. 60 rated) and compared 4 plunger suppliers.
- Modeled a multi-aisle warehouse robot base in SolidWorks, built on the company's existing track-based locomotion system.

Mechanical Engineer - Co-op | Hyderabad Batteries Limited | HYD, IND (Industry sponsored) **Oct 2021 - May 2022**

- Developed the Z-axis for a 3-axis Cartesian pick-and-place robot, delivered as a three-stage belt-driven design with released part drawings, by iterating concepts and modeling in SolidWorks/Onshape.
- Selected actuators, guides, and vacuum end-effector hardware from calculated requirements, producing a complete purchased-parts list reviewed with the HBL engineering team.
- Reduced component variation by 7% and procurement and assembly effort by 10%, as measured across 130 BOM line items, by consolidating duplicate wiring, connectors, and accessories.

RESEARCH EXPERIENCE

Northeastern University | DC Motor Control System **Sep 2025 - Dec 2025**

- Achieved ±2° positioning accuracy by building a PID speed-position controller in MATLAB/Simulink with Arduino, H-bridge, and encoder, and tuning Simscape models against test data.

Northeastern University | 2D Submarine Control Project **Mar 2025 - Apr 2025**

- Designed three MATLAB controllers (state-feedback, observer, PID) targeting 1% overshoot and 60 s settling for a simulated 10,000 kg vehicle; found designs exceeded motor thrust and recommended saturation modeling, pole changes, and larger thrusters.

Asia Pacific Robot Contest (ABU ROBOCON) | Mechanical Analyst - CAE Mechanical Head **Jun 2019 - Jun 2022**

- Co-developed two robots per season (20–25 kg limit) on omni and swerve-drive chassis, designing chassis, arms, and shooters and fixing hardware on-site; placed 12th (2020) and 15th (2021-2022) nationally.
- Trained 50+ students on CAD by leading an Onshape workshop.

SKILLS AND INTERESTS

CAD/CAE & Software: SolidWorks, Fusion 360, Onshape, MeshLab, FreeCAD, AutoCAD, ANSYS

Design and Documentation: GD&T, ASME/ANSI Y14.5, DFM/DFA, BOMs/ECOs, SAP/ECTR (PLM), stack-up analysis, CAPA (ISO 9001)

Prototyping : Additive manufacturing, CNC machining, laser cutting, sheet metal, soldering

Programming Languages & controllers: MATLAB/Simulink, Simscape, Python, C/Embedded C, PID and LQR, Arduino